

README

Spatium Computationis v3 Preprint Series Packet

Authority state: INTERNAL_RESEARCH

Source artifact: user_files/01-Pasted-text.txt

Repository inspected: <https://github.com/FreddyCreates/pegasus-battleops>

Inspected commit: a756fa39e39342dc2443c6c7d9335f4c9d68455c

Source title: Spatium Computationis v3: Mini-Verse Engines & Autonomous Intelligence Battleground Architecture

Prepared on: 2026-06-08

Intake Triage

- Work object: paper_draft, with secondary theorem_candidate, protocol_candidate, ip_claim, and architecture_note material.
- Task goal: packet_build for a publication-family preprint series.
- Claim class: primarily C3 hypothesis, C4 strategic thesis, C8 protocol candidate, C9 theorem candidate, and C10 public-safe educational claim. Selected architecture and deployment-surface statements are now repo-backed at the inspected commit.
- Proof posture: partial_repo_verified; the repository verifies the FastAPI app, pipeline endpoints, defense/honeypot architecture, platform router, tick-based 3D simulation, Docker/compose deployment scaffolding, and GitHub Actions test/build workflow. The advanced physics, Riemannian geometry, mini-verse cosmology, and quantitative validation claims from the pasted paper were not located in the inspected repo surface.
- Release boundary: evidence_needed_before_release with ip_review_first for architecture-sensitive agent/battleground mechanisms.
- Primary route: packet_build followed by claim hardening and hash/notary preparation.

Why This Route

- The source packet is broad enough to produce several distinct papers, but it mixes mathematical foundations, implementation claims, architectural claims, and strategic system framing.
- The strongest publication path is a 4-paper series, because each major contribution can carry its own thesis, evidence burden, and review audience.
- The largest proof gap is now contribution separation: the repository backs the operational battleground/simulation app, but the advanced physics/geometry/mini-verse papers still need file-level evidence or should be reframed as theory/design papers.
- Public release should avoid unverified production claims and should abstract sensitive battleground/agent-routing details until IP and release review are complete.

Packet Contents

- `series_blueprint.md`: preprint series structure, shared lineage, publication order, and overlap controls.
- `paper_01_physics_engine.md`: numerical mechanics and conservation-law paper.
- `paper_02_geometry_engine.md`: Riemannian geometry and geodesic computation paper.
- `paper_03_miniverse_cosmology.md`: mini-verse topology, cosmological evolution, and inter-verse boundary paper.
- `paper_04_agent_spacetime_architecture.md`: agent embedding and battleground architecture paper.
- `claims_evidence_matrix.md`: bounded claim classes, evidence state, gaps, and safer wording.
- `overlap_minimization_map.md`: scope rules to prevent repetition between papers.
- `shared_lineage.md`: common source lineage and derivative-policy record.
- `repo_evidence_map.md`: file-backed evidence map from the public repository.
- `hash_manifest.json`: local SHA-256 manifest for packet files.

series_blueprint

Preprint Series Blueprint

Series title: **Spatium Computationis v3: Physical Substrates for Autonomous Intelligence**

Series authority state: INTERNAL_RESEARCH

Publication posture: partial_repo_verified and needs citation hardening

Repository inspected: <https://github.com/FreddyCreates/pegasus-battleops>

Inspected commit: a756fa39e39342dc2443c6c7d9335f4c9d68455c

Repository verification state: repo_verified_architecture_partial_engine_gap

Series Thesis

Spatium Computationis v3 proposes that autonomous intelligence systems can be modeled as agents embedded in a physically constrained computational substrate, where simulation dynamics, differential geometry, cosmological sandboxing, and spatial agent routing become separable but connected layers of system design.

Recommended Series Structure

Paper 1: Numerical Mechanics Substrate

Title: **A Symplectic Physics Substrate for Autonomous Agent Simulation**

Distinct thesis: A physics substrate using Velocity Verlet integration, N-body interactions, relativistic speed bounds, elastic collisions, and conservation diagnostics can provide a bounded numerical environment for modeling agents as physical entities.

Primary contribution: numerical mechanics and validation framework.

Do not cover in depth: curvature tensors, mini-verse cosmology, agent battleground taxonomy.

Paper 2: Riemannian Geometry Engine

Title: **Numerical Riemannian Geometry for Runtime-Modifiable Computational Spacetimes**

Distinct thesis: A metric-function-first geometry engine can compute Christoffel symbols, geodesics, curvature tensors, horizons, and parallel transport numerically, enabling runtime-modifiable spacetime models without requiring closed-form symbolic derivations.

Primary contribution: computational differential geometry engine.

Do not cover in depth: N-body force laws, cosmological topology catalog, threat-routing architecture.

Paper 3: Mini-Verse Cosmology and Topological Sandboxes

Title: **Mini-Verse Sandboxes: Configurable Cosmological Substrates for Intelligence Experiments**

Distinct thesis: Isolated mini-verses with configurable constants, topologies, Friedmann evolution, entropy accounting, and bounded inter-verse channels can serve as experimental substrates for studying agent behavior under varied physical laws.

Primary contribution: universe-level sandbox architecture.

Do not cover in depth: integrator derivations, full curvature implementation, 21-agent spatial taxonomy.

Paper 4: Agent Embedding and Battleground Architecture

Title: **Embedding Autonomous Agents in Simulated Spacetime: A Battleground Architecture for Routing, Isolation, and Coordination**

Distinct thesis: Spatially embedding agents and zones inside a simulated substrate allows routing, quarantine, coordination, and influence to be expressed as geometric and physical relationships rather than only abstract software calls.

Primary contribution: autonomous agent architecture and spatial operational model.

Do not cover in depth: full numerical physics validation, tensor derivations, cosmological Friedmann analysis.

Publication Order

1. Paper 1 establishes the numerical substrate and validation norms.
2. Paper 2 extends the substrate into geometric computation.
3. Paper 3 packages geometry and physics into isolated universe-like experimental domains.
4. Paper 4 applies the substrate to autonomous intelligence architecture.

Shared Lineage Rule

Each paper should cite the canonical packet as the internal source lineage but should publish only its own contribution surface. Shared introduction text should be limited to one short paragraph defining Spatium Computationis and one paragraph stating the paper's position in the series.

Public/Private Boundary

Public-safe:

- Mathematical definitions of metrics, geodesics, conservation laws, and cosmological equations.
- General claim that the architecture proposes physically grounded agent simulation.
- High-level mini-verse taxonomy and sandbox framing.

Needs evidence before public promotion:

- Production-grade implementation claims.
- Quantitative validation thresholds such as energy error below 10^{-6} .
- Performance claims such as suitability for $N < 10000$.

IP/release review first:

- Detailed battleground routing logic.
- Security-oriented tiering and adversary-lab semantics.
- Exact agent placements if treated as strategic operational architecture.

Series-Level Evidence Needed

- Repository snapshot or release archive from `FreddyCreates/pegasus-battleops`.
- File-level evidence for Velocity Verlet/N-body physics, Riemannian geometry, and mini-verse cosmology if those remain paper claims.
- Commit hash matching the described implementation.
- Test logs for conservation laws, geodesic norm preservation, horizon detection, and topology behaviors.
- Benchmark table for N-body scaling, geodesic computation, curvature computation, and Web-Socket simulation tick behavior.
- Reproducibility script or notebook for at least one figure/table per paper.
- Citation pass for differential geometry, symplectic integration, cosmology, agent-based modeling, and cyber/defense architecture analogies.

paper_01_physics_engine

Paper 1 Brief: A Symplectic Physics Substrate for Autonomous Agent Simulation

Authority state: INTERNAL_RESEARCH

Claim boundary: `partial_repo_verified`; current repo verifies a tick-based 3D simulation layer, while Velocity Verlet/N-body/relativistic claims remain evidence-gated.

Abstract

This paper presents the numerical mechanics layer of *Spatium Computationis v3*, a proposed substrate for modeling autonomous agents as physically situated entities. The paper focuses on Velocity Verlet integration, pairwise gravitational and electromagnetic interactions, relativistic speed bounds, elastic collision handling, and conservation-law diagnostics. Its central claim is not that the system is fully validated in the supplied material, but that a symplectic mechanics layer provides a disciplined foundation for agent simulation when paired with measurable energy, momentum, angular-momentum, and causality checks. The paper defines the substrate, identifies its numerical invariants, and specifies the evidence required for public validation.

Distinct Thesis

A symplectic physics engine can serve as the first layer of a physically grounded autonomous-agent substrate by making agent interaction, influence, collision, and propagation constraints measurable through numerical conservation diagnostics.

Contribution Surface

- Velocity Verlet integration as the substrate's default time-stepping method.
- N-body force modeling with gravitational and Coulomb terms.
- Relativistic correction as a speed-bound mechanism.
- Elastic collision resolution for simulated entities.
- Conservation-law telemetry as runtime quality control.

Bounded Claims

Claim	Class	Proof posture	Public wording
Velocity Verlet is appropriate for long-running Hamiltonian-style simulation because its energy error is bounded under standard assumptions.	C10 / C9	Supported by known numerical analysis; needs citation pass.	"Velocity Verlet is a standard symplectic method often used when bounded long-term energy behavior is desirable."
The packet's physics engine implements N-body gravitational and electromagnetic interactions.	Potential C1	Not located in inspected repo surface.	"The canonical design targets N-body gravitational and electromagnetic interactions; the inspected repo currently verifies a tick-based 3D entity simulation layer."
Relativistic corrections prevent simulated particles from exceeding c.	C3 / potential C1	Needs implementation and test evidence.	"The design includes a relativistic speed-bound mechanism intended to constrain propagation speed."
Conservation diagnostics validate simulation quality.	C3 / C4	Methodologically plausible; thresholds need logs.	"Conservation diagnostics are proposed as quality indicators for simulation runs."

Proposed Outline

1. Introduction: physical substrate for agent simulation.
2. Numerical mechanics requirements.
3. Velocity Verlet integration and error behavior.
4. Pairwise interaction model.
5. Relativistic speed-bound policy.
6. Collision model and conservation assumptions.
7. Conservation-law telemetry.
8. Validation plan and expected diagnostics.
9. Limits and relation to later series papers.

Required Evidence Before Preprint Release

- Code references for integrator, force computation, collision resolution, and telemetry.
- Current repo-backed references: `spatium-computationis/simulation/engine.py`, `spatium-computationis/simulation/world.py`, and `spatium-computationis/simulation/routes.py` for tick-based 3D simulation.
- Energy, momentum, and angular-momentum drift plots across representative runs.
- Parameter table for time step, softening length, masses, charges, and units.
- Comparison against Euler or RK methods if claiming superior long-term behavior.
- Benchmark confirming scaling and practical particle limits.

Minimized Overlap Rule

This paper may mention agents only as physical entities requiring a numerical substrate. It should not describe the 21-agent architecture, mini-verse topology catalog, or full differential-geometry machinery except as future layers.

paper_02_geometry_engine

Paper 2 Brief: Numerical Riemannian Geometry for Runtime-Modifiable Computational Spacetimes

Authority state: INTERNAL_RESEARCH

Claim boundary: C3, C9, C10; implementation claims held as evidence_needed_before_release because matching geometry-engine files were not located in the inspected repo surface.

Abstract

This paper isolates the Riemannian geometry layer of Spatium Computationis v3. It frames the system as a metric-function-first engine for numerical differential geometry: given a smooth metric function, the engine estimates Christoffel symbols, integrates geodesics, computes curvature quantities, detects horizons, and supports parallel transport. The paper's distinct contribution is the runtime-modifiable geometry model, not the physics force layer or the full agent architecture. Public claims should be limited to mathematical formulation and proposed computational design until linked implementation evidence and numerical validation traces are attached.

Distinct Thesis

A metric-function-first geometry engine can make curved computational spacetimes operational for autonomous systems by numerically deriving connection, geodesic, curvature, horizon, and transport behavior from runtime-defined metrics.

Contribution Surface

- Metric tensor and finite-difference derivative model.
- Christoffel-symbol computation from arbitrary metric functions.
- RK4 geodesic integration and tangent norm preservation.
- Riemann, Ricci, scalar curvature, Einstein tensor, and Kretschmann scalar analysis.
- Horizon detection via metric-sign behavior.
- Parallel transport as a computational measure of holonomy.

Bounded Claims

Claim	Class	Proof posture	Public wording
The geometry layer computes Christoffel symbols numerically using central finite differences.	Potential C1	Not located in inspected repo surface.	"The design specifies central finite differences for estimating metric derivatives; implementation verification requires geometry-engine files."
Geodesics are integrated using RK4 with adaptive step control.	Potential C1	Not located in inspected repo surface.	"The design proposes RK4 geodesic integration; adaptive-step details should be tied to implementation evidence."
Curvature analysis includes Riemann, Ricci, scalar curvature, Einstein tensor, and Kretschmann scalar.	Potential C1	Not located in inspected repo surface.	"The planned curvature API includes standard curvature invariants and tensors; public claims should cite implementation and tests once available."
Runtime-modifiable spacetime works for any smooth metric.	C9	Mathematically plausible but too broad without constraints.	"The approach is intended for smooth metric functions within numerical stability limits."

Proposed Outline

1. Introduction: why geometry should be separated from mechanics.
2. Metric-function interface.
3. Numerical connection coefficients.
4. Geodesic integration pipeline.
5. Curvature computation and invariant diagnostics.
6. Horizon detection and coordinate caveats.
7. Parallel transport and holonomy.
8. Validation cases: Minkowski, Schwarzschild, FLRW, AdS.
9. Limitations: finite differences, singularities, coordinate dependence, computational cost.

Required Evidence Before Preprint Release

- Unit tests comparing known analytic Christoffel symbols and curvature values.
- Geodesic norm preservation plots for benchmark metrics.
- Horizon detection tests for Schwarzschild radius.
- Error sensitivity analysis for finite-difference step size.
- Complexity and runtime benchmark for curvature computation in 4D.

Minimized Overlap Rule

This paper may reference the physics engine only as a separate substrate layer. It should not repeat N-body equations except to clarify that force-based particle dynamics and geometry-based geodesics are distinct computational regimes.

paper_03_miniverse_cosmology

Paper 3 Brief: Mini-Verse Sandboxes as Configurable Cosmological Substrates

Authority state: INTERNAL_RESEARCH

Claim boundary: C3, C4, C8, C9, C10; implementation claims held as evidence_needed_before_release because matching mini-verse files were not located in the inspected repo surface.

Abstract

This paper turns the mini-verse material in the canonical packet into a standalone preprint on configurable cosmological sandboxes for autonomous-intelligence experiments. It treats each mini-verse as an isolated universe instance with its own constants, topology, particle population, cosmological evolution, entropy accounting, and bounded inter-verse communication channels. The paper's contribution is a sandbox architecture for varying physical assumptions, not a proof that agent intelligence improves under those assumptions. Implementation and experimental claims should remain bounded until reproducible runs compare agent behavior across topologies and constants.

Distinct Thesis

Configurable mini-verses can serve as experimental substrates for studying how autonomous-agent behavior changes under different physical constants, topological constraints, cosmological phases, and inter-domain boundary conditions.

Contribution Surface

- Mini-verse object model: constants, topology, geometry, particles, and evolution state.
- Topology catalog: flat infinite, flat periodic, spherical, hyperbolic, de Sitter, and anti-de Sitter.
- Friedmann evolution as universe-level simulation policy.
- Thermodynamic arrow of time using particle and horizon entropy accounting.
- Inter-verse boundary channels: wormholes, entanglement-like information channels, and membranes.

Bounded Claims

Claim	Class	Proof posture	Public wording
Mini-verses can be instantiated with configurable physical constants and topologies.	Potential C1 / C8	Not located in inspected repo surface.	"The research architecture defines mini-verse instances with configurable constants and topology choices; implementation claims should cite matching repository files once located."
Friedmann equations govern cosmological evolution inside each mini-verse.	C10 / potential C1	Equations are standard; implementation needs evidence.	"The proposed evolution model uses Friedmann-style equations for scale-factor dynamics."
Entropy tracking provides an arrow of time.	C3 / C9	Needs model specification and validation.	"Entropy accounting is proposed as a runtime indicator of thermodynamic directionality."
Inter-verse channels enforce causality through bandwidth limits.	C3 / C8	Needs formal semantics and tests.	"Inter-verse channels are specified with bandwidth limits intended to constrain transfer."

Proposed Outline

1. Introduction: why isolate cosmological sandboxing from engine internals.
2. Mini-verse object model and lifecycle.
3. Topology catalog and boundary behavior.
4. Friedmann evolution and phase classification.
5. Entropy accounting and thermodynamic direction.
6. Inter-verse connection types and boundary constraints.
7. Experimental designs for agent behavior under varied constants/topologies.
8. Validation requirements and failure modes.
9. Relationship to physics, geometry, and agent architecture papers.

Required Evidence Before Preprint Release

- Formal mini-verse schema or class interface.
- Reproducible examples for at least three topologies.
- Scale-factor evolution traces for radiation, matter, dark-energy, and contracting phases.
- Entropy tracking definition with numeric examples.
- Transfer tests showing bandwidth-limited inter-verse channels.

Minimized Overlap Rule

This paper should not re-derive Christoffel symbols or Velocity Verlet mechanics beyond short citations to Papers 1 and 2. Its novelty is the universe-level composition and experiment design layer.

paper_04_agent_spacetime_architecture

Paper 4 Brief: Embedding Autonomous Agents in Simulated Spacetime

Authority state: INTERNAL_RESEARCH

Claim boundary: repo_verified_architecture; sensitive security framing still requires IP/release review.

Abstract

This paper converts the organism-agent and battleground material in the canonical packet into a bounded architecture preprint. It studies the design pattern of embedding autonomous agents and operational zones inside a simulated spacetime so that routing, isolation, influence, communication, and quarantine can be represented spatially. Because this thread contains security-oriented tiering, adversary-lab semantics, and potentially strategic routing mechanisms, publication should remain high-level unless Medina approves the release boundary. The paper should present the architecture as a design proposal and experimental framework, not as a verified production security system.

Distinct Thesis

Spatial embedding offers a design language for autonomous-agent coordination in which operational roles, trust boundaries, influence, and quarantine can be modeled as geometric regions and physical interaction rules.

Contribution Surface

- Three-tier signal classification: cooperative, hostile, shadow.
- Agent placement as position-bearing entities in a simulated substrate.
- Zone definitions as bounded geometric regions with semantic roles.
- Influence and communication modeled through force analogies.
- Architecture bridge from physical substrate to operational intelligence routing.

Bounded Claims

Claim	Class	Proof posture	Public wording
The architecture embeds agents as physical entities in simulated 3D space.	C5 / partial C1	Repo-backed through simulation/engine.py, simulation/world.py, and simulation/routes.py; full 21-agent physical placement not all visible in resident simulation list.	"The implementation embeds resident agents, threats, signals, honeypots, data packets, and defensive nodes in a tick-based 3D operational space."
Three-tier routing separates cooperative, hostile, and unknown signals.	C5 / C8	Repo-backed as README architecture; release-sensitive.	"The design documents a tiered routing model for cooperative, adversarial, and unknown inputs."
Gravitational attraction models influence or authority.	C4 / C9	Metaphorical/protocol claim; not empirical proof.	"The architecture uses physical analogies, such as attraction, to model influence relationships."
The system is production-oriented.	Potential C1	Repo-backed deployment scaffolding; live production operation not proven.	"The repository contains production-oriented deployment scaffolding, including Docker, compose, and CI build/test workflow."

Proposed Outline

1. Introduction: agent systems as spatial architectures.
2. Separation between physics substrate and operational semantics.
3. Signal classification and routing model.
4. Agent embedding: positions, zones, and interaction surfaces.
5. Zone geometry as policy boundary.
6. Influence, communication, quarantine, and isolation as spatial operations.
7. Release-safe architectural diagrams.
8. Evaluation plan: routing correctness, containment, latency, and failure modes.
9. IP-sensitive details and public/private boundary.

Required Evidence Before Preprint Release

- Agent registry or architecture diagram approved for release.
- Current repo-backed references: README.md, spatium-computationis/main.py, spatium-computationis/simulation/engine.py, spatium-computationis/simulation/world.py, spatium-computationis/defense/honeypot/routes.py, spatium-computationis/platform_router.py, Dockerfile, docker-compose.yml, and .github/workflows/deploy-app.yml.
- Non-sensitive pseudocode for tier classification and routing.
- Simulation traces showing movement, routing, quarantine, or zone transitions.
- Threat-model document and containment criteria.

- IP review deciding which names, zones, coordinates, and mechanisms may be public.

Minimized Overlap Rule

This paper should cite Papers 1-3 as substrate layers and focus on operational semantics. It should not restate the mathematical derivations or cosmology machinery except to explain how they constrain the agent architecture.

claims_evidence_matrix

Claims and Evidence Matrix

Authority state: CLAIM_HARDENED_DRAFT

Overall proof posture: partial_repo_verified

Reason: The uploaded packet contains mathematical exposition and architecture description, and the public repository was inspected at commit a756fa39e39342dc2443c6c7d9335f4c9d68455c. The repository verifies the operational app, defense/platform architecture, tick-based 3D simulation, and deployment scaffolding. Advanced physics, Riemannian geometry, mini-verse cosmology, and numerical validation claims remain evidence-gated because matching implementation files were not located in the inspected surface.

Claim Review Table

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-001	Series	Spatium Computationis models autonomous agents as physical entities in simulated spacetime.	C4, potential C1 for current 3D simulation substrate	Repo-backed for tick-based 3D entity simulation; advanced spacetime claims remain design/theory.	Medium	Cite simulation/world.py, simulation/engine.py, simulation/routes.py; add formal model for stronger claims.	“Spatium Computationis implements a tick-based 3D simulation substrate for agents, signals, threats, zones, and data packets; broader spacetime claims remain part of the research framing.”

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-002	Paper 1	The physics engine implements Velocity Verlet integration.	Potential C1	Not located in inspected repo surface.	High	Specific physics-engine file, tests, and run logs.	"The canonical paper proposes a Velocity Verlet physics layer; the inspected repo currently verifies a tick-based 3D simulation layer."
C-003	Paper 1	Verlet integration yields bounded energy error for Hamiltonian systems.	C10, C9	Externally supportable with citations.	Low	Cite Verlet and geometric numerical integration sources.	"Under standard assumptions, symplectic methods such as Velocity Verlet are used for bounded long-term energy behavior."
C-004	Paper 1	The system enforces a hard speed bound at c .	C3, potential C1	Not located in inspected repo surface.	High	Relativistic update tests and edge-case traces.	"The research design includes a relativistic correction intended to bound simulated propagation speed."
C-005	Paper 1	Conservation error remains below 10^{-6} for typical time steps.	Potential C1	Not located in inspected repo surface.	High	Run logs, plots, parameter definitions.	"Validation should report conservation error under specified time-step and parameter regimes."

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-006	Paper 2	Geometry quantities are computed numerically from metric functions.	Potential C1, C9	Not located in inspected repo surface.	High	Code and benchmark metrics.	“The geometry design uses metric functions as the interface for numerical differential geometry.”
C-007	Paper 2	The engine works for any smooth metric.	C9	Overbroad without numerical constraints.	High	Stability domain, convergence tests, singularity handling.	“The approach targets smooth metric functions within defined numerical stability limits.”
C-008	Paper 2	Horizon detection can be performed by scanning sign changes in g_{tt} .	C10, potential C1	Conceptually supported but coordinate-sensitive.	Medium	Benchmark metrics and caveat discussion.	“For suitable coordinates, horizon candidates may be detected by monitoring metric component behavior such as sign changes in g_{tt} .”
C-009	Paper 3	Mini-verses can use configurable constants and topologies.	C8, potential C1	Not located in inspected repo surface.	High	Schema, examples, tests.	“The research architecture defines mini-verse instances with configurable constants and topology choices.”

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-010	Paper 3	Entropy tracking provides a thermodynamic arrow of time.	C3, C9	Needs formalization.	Medium	Entropy model and numeric examples.	"Entropy accounting is proposed as an indicator of thermodynamic directionality."
C-011	Paper 3	Inter-verse channels enforce causality through bandwidth limits.	C8, C3	Not located in inspected repo surface.	High	Transfer model, tests, causal proof sketch.	"The architecture specifies bandwidth-limited inter-verse channels intended to constrain transfer."
C-012	Paper 4	Agents interact through physical forces where attraction models influence and electromagnetic forces model communication.	C4, C9	Metaphorical/prototypical claim.	Medium	Formal mapping from physical variables to operational semantics.	"The architecture uses physical analogies to represent influence and communication relationships."
C-013	Paper 4	The battleground architecture contains cooperative, hostile, and shadow routing tiers.	C5, C8, partial C1 for documented routes	Repo-backed architecture and route surfaces; release-gated for sensitive details.	Medium	Cite README.md, main.py, defense/honeypot/router.py, platform_router.py.	"A tiered battleground architecture is documented and partially implemented through defense, honeypot, platform, and simulation routes."

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-014	Series	The system is production-grade.	Potential C1	Deployment scaffolding is repo-backed; production operation still needs live deployment/CI run evidence.	Medium	Workflow run status, release artifact, deployment URL, operational logs.	“The repository contains production-oriented deployment scaffolding, including Docker, compose, and CI build/test workflow; live production status requires runtime evidence.”

Citation and Evidence Gaps

Missing external citations:

- Symplectic integration and long-term energy behavior.
- Numerical differential geometry and finite-difference curvature computation.
- General relativity references for benchmark metrics and invariants.
- Cosmological evolution and entropy references.
- Agent-based modeling, cyber range, sandbox, and immune-system-inspired defense literature.

Repository evidence status:

- Repository identified by operator: <https://github.com/FreddyCreates/pegasus-battleops>.
- Runtime access status: GitHub connector succeeded; shell clone remained blocked by the container network.
- Commit inspected: a756fa39e39342dc2443c6c7d9335f4c9d68455c.
- Promotion rule: implementation claims may be upgraded only when tied to specific files, commits, tests, and logs.

Missing internal evidence still needing inspection:

- File-level implementation evidence for advanced physics, geometry, and mini-verse claims.
- Test suite contents and actual output logs.
- Benchmark results.
- Diagrams or screenshots of the runtime/visualization if visual claims are retained.

Missing experiments or validations:

- Energy/momentum/angular momentum conservation traces.
- Geodesic norm preservation and analytic comparison tests.

- Horizon detection benchmark.
- Mini-verse topology behavior examples.
- Agent routing and containment simulation traces.

Claims to remove or downgrade before public release:

- “production-grade” should be narrowed to “production-oriented deployment scaffolding” unless live deployment, passing CI run, or operational evidence is attached.
- “all implementations use real SI units” unless unit tests or code evidence are attached.
- “validated conservation laws” unless actual validation output is included.
- “proper differential geometric constructions” should be reframed as “standard mathematical constructions are specified” unless implementation tests are attached.

Next Proof Move

Use the repo-backed architecture and simulation evidence to strengthen Paper 4 immediately, and either locate the missing advanced engine files or reframe Papers 1-3 as staged theory/design papers aligned with the current tick-based 3D simulation substrate.

overlap_minimization_map

Overlap Minimization Map

Goal: preserve shared lineage while preventing each preprint from repeating the same mathematics, architecture claims, or implementation assertions.

Shared Text Budget

Each paper may reuse:

- One paragraph defining Spatium Computationis v3.
- One paragraph placing the paper within the four-part series.
- One short source-lineage note.
- One compact limitations paragraph referencing evidence gaps.

Each paper should avoid:

- Copying the full original abstract.
- Repeating all four canonical contributions.
- Reusing the same conclusion claims.
- Restating all equations from other papers.

Paper-Specific Ownership

Material	Owner paper	Allowed cross-reference
Velocity Verlet equations	Paper 1	Paper 4 may cite as substrate, no derivation.
N-body gravitational and Coulomb force laws	Paper 1	Paper 3 may mention particle dynamics only.
Relativistic speed bound	Paper 1	Paper 4 may cite as causality constraint.
Conservation-law diagnostics	Paper 1	Papers 2-4 may cite validation norm.
Metric tensor and Christoffel symbols	Paper 2	Paper 3 may cite metric substrate.
Geodesic equation and RK4 integration	Paper 2	Paper 4 may cite geometric routing conceptually.
Curvature tensors and invariants	Paper 2	Paper 3 may mention topology/curvature at high level.
Horizon detection and parallel transport	Paper 2	Paper 3 may cite horizon entropy only.

Material	Owner paper	Allowed cross-reference
Topology catalog	Paper 3	Paper 2 may list benchmark metrics without sandbox framing.
Friedmann evolution	Paper 3	Paper 2 may mention FLRW as a metric example only.
Entropy and inter-verse channels	Paper 3	Paper 4 may cite sandbox boundaries only.
Three-tier signal classification	Paper 4	No other paper should include details.
Agent coordinates and zones	Paper 4	Papers 1-3 should not include them.
Security/battleground semantics	Paper 4	No other paper should include details.

Abstract Differentiation

Paper 1 abstract vocabulary:

- numerical mechanics
- symplectic integration
- conservation diagnostics
- force-mediated simulation

Paper 2 abstract vocabulary:

- metric functions
- numerical differential geometry
- geodesic computation
- curvature analysis

Paper 3 abstract vocabulary:

- mini-verse
- configurable constants
- topology
- Friedmann evolution
- entropy and inter-verse boundaries

Paper 4 abstract vocabulary:

- autonomous agents
- spatial embedding
- routing
- quarantine
- operational zones

Conclusion Differentiation

- Paper 1 concludes with a validation-first mechanics substrate.
- Paper 2 concludes with geometry as a runtime-modifiable computational layer.

- Paper 3 concludes with sandbox design and experiment families.
- Paper 4 concludes with spatial architecture for agent coordination and release-gated security semantics.

Shared Lineage Citation Pattern

Recommended internal citation note:

This paper derives from the canonical Spatium Computationis v3 packet, version 3.0, prepared in June 2026. It isolates one contribution surface from the broader packet to reduce overlap and clarify evidence requirements.

Anti-Overlap Reviewer Checklist

- Does the paper make one primary contribution?
- Are equations copied only when they are owned by the paper?
- Are implementation claims stated only where evidence is expected?
- Does the abstract avoid listing all four series contributions?
- Does the conclusion avoid claiming the whole system is validated?
- Are private or security-sensitive details omitted unless this is Paper 4 and release-approved?

repo_evidence_map

Repository Evidence Map

Authority state: REPO_EVIDENCE_PARTIAL

Repository: <https://github.com/FreddyCreates/pegasus-battleops>

Commit inspected: a756fa39e39342dc2443c6c7d9335f4c9d68455c

Access route: GitHub connector. Local git clone remained blocked by the container network, but connector file fetches succeeded.

Verified Repository Surfaces

Evidence target	Repository file	Verification state	Notes
Public repo identity	Repository metadata	verified	Repo is public, default branch main, size reported as 58490, clone URL available. Describes Spatium Computationis as AI intelligence battleground, organism defense, autonomous agent ecosystem, with 21 agents and 53 protocols. Documents FastAPI app, 5-protocol intelligence pipeline, 7-agent operational layer, API endpoints, and Nova Sovereign runtime configuration.
Top-level system description	README.md	repo_verified_architecture	
Furniture/interiors intelligence layer	spatium-computationis/README.md	repo_verified_architecture	

Evidence target	Repository file	Verification state	Notes
FastAPI application endpoint	spatium-computationis/main.py	repo_verified_implementation_surface	Defines FastAPI app version 0.5.0, mounts static files, includes defense, marketing, platform, simulation, and frontend routers, and exposes /, /ingest, /field-update, /memory/recall, direct agent endpoints, and defense webhook/WebSocket endpoints.
Pipeline behavior	spatium-computationis/main.py	repo_verified_implementation_surface	ingest executes ingressus -> compressio -> ordinatio -> actio; /field-update executes Reductus feedback processing.
Deployment scaffold	Dockerfile and docker-compose.yml	repo_verified_deployment_scaffold	Dockerfile builds Python 3.12 app and runs uvicorn; compose exposes port 8000, passes Nova env vars, and defines a healthcheck.
CI/deployment workflow	.github/workflows/deploy-app.yml	repo_verified_ci_scaffold	Workflow installs dependencies, runs pytest tests/, runs tests/test_maesi_smoke.py, then builds and pushes a Docker image to GHCR.
Honeypot defense endpoints	spatium-computationis/defense/honeypot/routes.py	repo_verified_implementation_surface	Defines fake .env, config, admin, WordPress, login, API, and service endpoints; collects fingerprints and records honeypot triggers.
Platform agent/task router	spatium-computationis/platform_router.py	repo_verified_implementation_surface	Exposes platform agent discovery, agent health, task queue, delegation, buses, taxis, and layer endpoints.
3D simulation API	spatium-computationis/simulation/routes.py	repo_verified_implementation_surface	Exposes /simulation/state, /stats, /start, /stop, /events, /zones, entity lookup, entity injection, and WebSocket streaming.
Tick-based simulation engine	spatium-computationis/simulation/engine.py	repo_verified_implementation_surface	Defines TICK_RATE = 20, resident agents, honeypot nodes, defense nodes, procedural spawning of signals/threats/data packets, subscriber broadcasting, injection, state, and stats APIs.

Evidence target	Repository file	Verification state	Notes
3D world model	spatium-computationis/simulation/world.py	repo_verified_implementation_surface	Defines Vector3, EntityType, EntityState, ZoneType, bounded zones, entities, movement toward targets, state serialization, deltas, and event logs.
Agent/glyph mapping	spatium-computationis/glyphs/glyph_map.py	repo_verified_implementation_surface	Maps operational glyphs to regions and agents, including defense, shadow, gatekeeper, adversary, and research agents.

Claims Upgraded By Repo Evidence

- The system has a public repository with a FastAPI application surface.
- The operational architecture includes organism defense, honeypots, platform routing, task queues, agent discovery, and direct agent endpoints.
- The repo contains a tick-based 3D simulation substrate with entities, zones, resident agents, procedural signal/threat/data-packet spawning, and WebSocket streaming.
- Deployment scaffolding exists through Docker, docker-compose, and a GitHub Actions workflow that runs tests and builds/pushes a container image.
- Paper 4 can now be treated as the strongest repo-backed paper in the series.

Claims Still Evidence-Gated

The following pasted-paper claims were not located in the inspected repository surface and should not be promoted to verified implementation claims yet:

- Velocity Verlet integration.
- N-body gravitational and electromagnetic interactions.
- Relativistic correction with hard speed bound at c .
- Conservation-law tracking with numerical error thresholds.
- Riemannian metric engine, Christoffel symbols, geodesic RK4 integration, curvature tensors, Einstein tensor, Kretschmann scalar, horizon detection, and parallel transport.
- Mini-verse constants, topology catalog implementation, Friedmann evolution, entropy accounting, and inter-verse channels.

Recommended Series Revision

- Keep Paper 4 as repo-backed architecture paper.
- Reframe Paper 1 as the current tick-based 3D simulation substrate unless matching physics-engine files are supplied.
- Reframe Papers 2 and 3 as theory/design roadmap papers unless geometry and mini-verse implementation files are located.
- Add a repo-backed appendix citing the inspected files and commit.

shared_lineage

Shared Lineage Record

Lineage schema: medina.preprint_series_lineage.v1

Prepared on: 2026-06-08

Prepared by: THESIS Alpha

Source Artifact

- Source path: user_files/01-Pasted-text.txt
- Repository inspected: <https://github.com/FreddyCreates/pegasus-battleops>
- Repository commit inspected: a756fa39 full hash retained in hash_manifest.json
- Repository access status in this runtime: GitHub connector accessible; shell clone blocked by container network
- Source title: Spatium Computationis v3: Mini-Verse Engines & Autonomous Intelligence Battleground Architecture
- Source author field: FreddyCreates
- Source version: 3.0
- Source date field: June 2026
- Source declared destination: A Zenodo Technical Paper

Derivative Series

Series title: Spatium Computationis v3: Physical Substrates for Autonomous Intelligence

Derivative papers:

1. A Symplectic Physics Substrate for Autonomous Agent Simulation
2. Numerical Riemannian Geometry for Runtime-Modifiable Computational Spacetimes
3. Mini-Verse Sandboxes: Configurable Cosmological Substrates for Intelligence Experiments
4. Embedding Autonomous Agents in Simulated Spacetime: A Battleground Architecture for Routing, Isolation, and Coordination

Authority State

- Source authority: DRAFT / INTERNAL_RESEARCH
- Derivative authority: INTERNAL_RESEARCH

- Claim-hardening state: CLAIM_HARDENED_DRAFT
- Release state: evidence_needed_before_release
- Notary state: local_hash_ready_after_manifest

Packet Backbone

- packet_id: SCV3-PREPRINT-SERIES-2026-06-08
- packet_class: publication_family_preprint_series
- packet_family: Publication
- source_artifact: user_files/01-Pasted-text.txt
- source_artifact_type: canonical_latex_paper_packet
- artifact_selection_basis: user_explicitly_uploaded_source
- authority_state: INTERNAL_RESEARCH
- evidence_state: partial_repo_verified
- claim_class_summary: C3, C4, C5, C8, C9, C10; potential C1 blocked pending evidence
- provenance_refs: uploaded_source_packet; FreddyCreates/pegasus-battleops@a756fa39
- hash_manifest_ref: hash_manifest.json
- policy_boundary: public release requires citation hardening, evidence attachment, and IP review for Paper 4
- derivative_policy: minimize overlap by assigning one contribution surface per paper
- review_state: operator_review_required
- promotion_state: not_public_safe_yet
- destination_surface: preprint_series / Zenodo-ready after proof and citation hardening
- operator_notes: Repository inspected. Architecture/simulation/deployment surfaces are repo-backed; advanced physics/geometry/mini-verse claims still need file-level evidence or reframing.

Memory Append Recommendation

Append a MEMORIA record noting that the canonical Spatium Computationis v3 packet was split into a four-paper preprint series with preserved source lineage, bounded claims, local hash-manifest preparation, and repository lineage inspected at FreddyCreates/pegasus-battleops@a756fa39. Promote only the architecture, simulation, and deployment claims verified in repo_evidence_map.md. Keep advanced physics, geometry, and mini-verse claims evidence-gated until matching files are located.